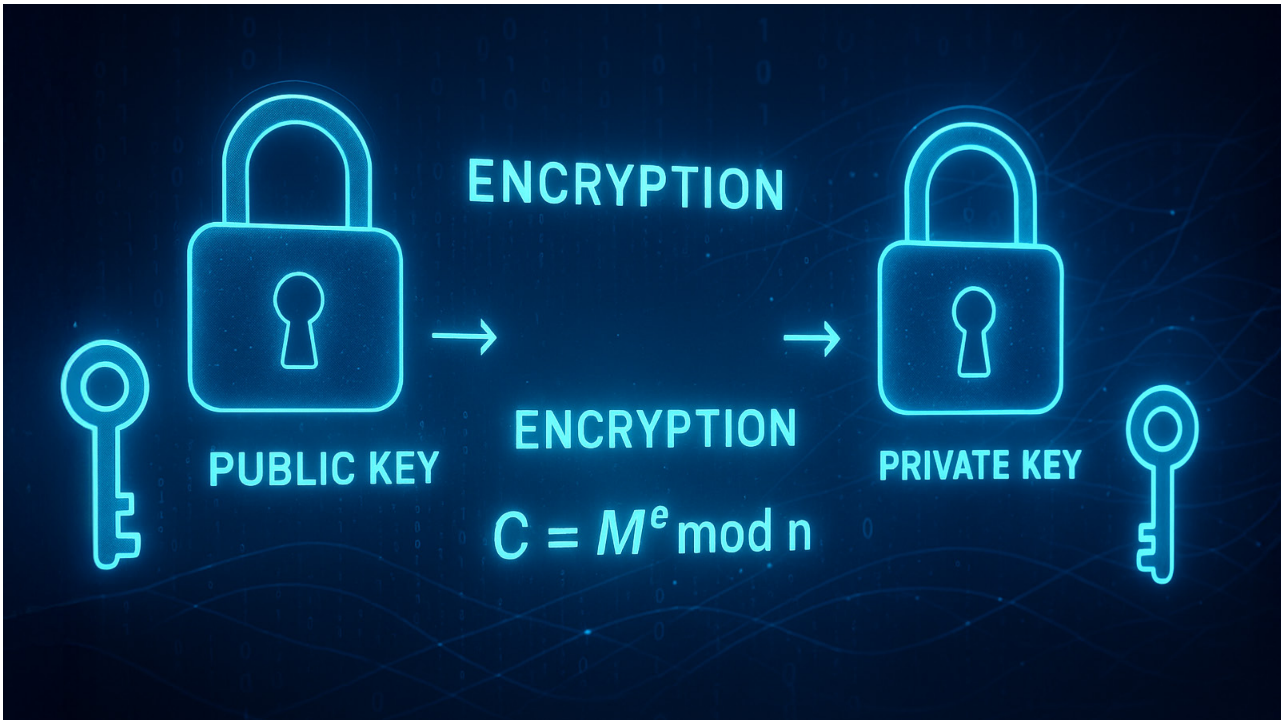
SageMath: Cracking the RSA Paradox

This eleventh part of the series on SageMath delves deeper into the RSA algorithm and demonstrates its implementation.



In the last article in this series on SageMath *(OSFY, August 2025),* we explored recent trends in quantum computing and how they are poised to impact cybersecurity in the near future. Our discussion also touched upon cryptanalysis and introduced the concept of public-key cryptography (PKC), along with a short program to help readers grasp the basic idea of PKC. As promised, this article will delve deeper into the RSA algorithm and demonstrate its implementation. However, before we proceed, it is important to address a foundational question. Throughout this series, we have seen how SageMath seamlessly integrates Python features. This may prompt some readers to wonder: Is SageMath simply Python with a few additional functions, merely masquerading as a distinct tool? The answer is a resounding 'No'. In the following paragraphs, we will explore why SageMath stands on its own as a powerful and independent mathematical software system—far more than just another dialect of Python.

To understand why SageMath is more than just Python with a mathematical flair, we need to look at what it offers under the hood. At its core, SageMath is a unified, Python-based interface to a vast ecosystem of mature, open source mathematical software. This includes powerful tools like GAP (for group theory), PARI/GP (for arithmetic geometry), Maxima (for symbolic computation), Singular (for algebraic geometry), and many others. What SageMath does remarkably well is integrate these diverse systems into a single, consistent environment. This allows users to access their immense power without needing to learn each system's individual syntax or quirks. Beyond this integration, SageMath also extends Python with its own rich set of native mathematical types and operations, providing a comprehensive and seamless experience for complex computations.

Figures 1 and 2 showcase five one-liners each from Python and SageMath, respectively, demonstrating their fundamental differences.